

How Engineering Intensity Shapes Flow-Path Structure:

Evaluating Width-Functions for Urban, Forest, and Agricultural Basins

Ori Clark GES 419

Introduction

Urbanization reshapes natural hydrologic systems by increasing the amount of impervious surfaces and stormwater features.

Pipes, inlets, and culverts are designed to protect infrastructure by quickly collecting and conveying runoff, and alter how water moves across a landscape.

Flow-path structure — the pattern of distances water travels before reaching an outlet .

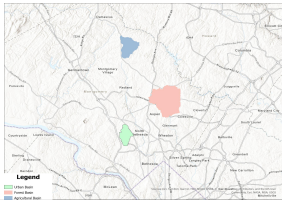
A **width function** summarizes how much contributing area exists at each flow-path distance and is useful for comparing hydrologic behavior across different landscapes.

Objective

Compare three watersheds (forested, agricultural, and urban) in Montgomery County, MD.

Determine whether basins with engineered infrastructure (urban) exhibit a **different flow length distribution** than more natural basins (forested and rural agricultural).

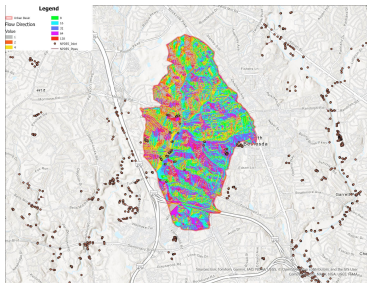
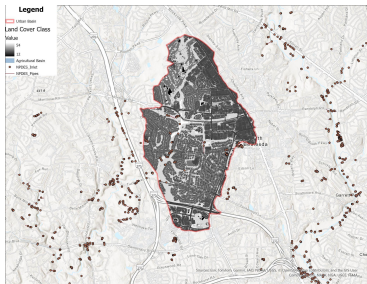
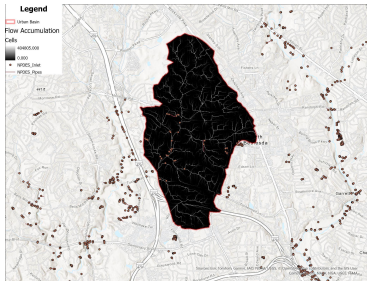
Evaluate peak flow path lengths, distribution spread, and skewness



Methods

A 10m DEM obtained and hydrologically conditioned.

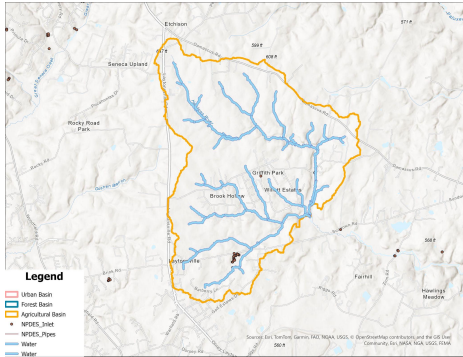
- **Basin Selection:** Identified by overlaying land cover and NPDES data
- **Watershed Delineation:** Derived flow direction and flow accumulation grids Pour points placed correct outlet.



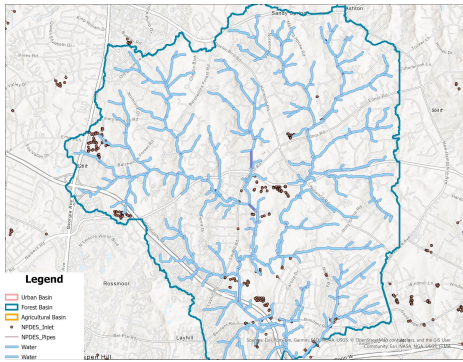
Width Function Analysis: Using the flow direction grid and outlet point, we calculated for the **flow length** for each cell in the basin.

Next, the **zonal histogram** tool or custom script tallied the count of cells (area) at specified distance intervals from the outlet.

Natural



Engineered



Results

Forested: Peaks at ~200.

- Contains more stormwater features than the rural basin.

Agricultural: Peaks at ~160.

- Has fewer stormwater features yet drains more efficiently

Urban: Peaks at ~100,

- Municipal stormwater network routes runoff through pipes and impervious surfaces to shorten residence time

